

Problem 4.1*

Let $J : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function. Show that if $x_0 < x_1 < x_2$ then

$$\frac{J(x_1) - J(x_0)}{x_1 - x_0} \leq \frac{J(x_2) - J(x_0)}{x_2 - x_0} \leq \frac{J(x_2) - J(x_1)}{x_2 - x_1}.$$

Solution.

Problem 4.2*

Let $J \in C^1(\mathbb{R})$.

- (a) Show that J is convex if and only if its derivative J' is monotone non-decreasing.
- (b) Show that if $r \in (1, \infty)$ then $x \rightarrow J(x) := |x|^r$ is convex and $J(x) \geq J(x_0) + J'(x_0)(x - x_0)$ for all $x, x_0 \in (0, \infty)$.

Solution.

Problem 4.3*

Let (Ω, Σ, μ) be an outer measure space with $\mu(\Omega) = 1$. Let $r > 1$ and set $\phi(t) = t^r$ for $t \geq 0$. Show that if $g : \Omega \rightarrow [0, \infty)$ is measurable then

$$\int_{\Omega} g^r(x) \mu(dx) \geq \left(\int_{\Omega} g(x) \mu(dx) \right)^r.$$

Solution.

Problem 4.4*

Let (Ω, Σ, μ) be an outer measure space with $\mu(\Omega) < \infty$. Let $f, f_1, f_2, \dots : \Omega \rightarrow \mathbb{R}$ be measurable functions on Ω and assume $f_j(x) \rightarrow f(x)$ as $j \rightarrow \infty$ for almost every x . Assume

$$\int_{\Omega} f^2(x) \mu(dx) < \infty \quad \text{and} \quad \int_{\Omega} f_j^2(x) \mu(dx) < 1 \quad \forall j \in \mathbb{N}.$$

Show that $\int_{\Omega} |f_j(x) - f(x)|^p \mu(dx) \rightarrow 0$ for every $p \in (0, 2)$. Construct a counterexample to show this can fail when $p = 2$.

Solution.

Problem 4.5*

Let μ be an outer measure on X , let $f : X \rightarrow [0, +\infty)$ be μ -measurable. For $t \geq 0$ we define

$$S_f(t) = \{x \in X : f(x) > t\}, \quad F_f(t) = \mu[S_f(t)].$$

- (a) Show that F_f is $\mathcal{L}^1$ -measurable.
 (b) Show that $\int_X f \, d\mu = \int_0^\infty F_f \, d\mathcal{L}^1$ (you could first check the case when f assumes finitely many values).

Solution.

Problem 4.6

Let μ be the Lebesgue measure on $(0, 1)$. Show that the parallelogram law fails on $L^1(\mu)$.

Solution.

Problem 4.7*

Let $1 \leq q < +\infty$. Assume that $f \in L^q(\Omega, d\mu) \cap L^\infty(\Omega, d\mu)$. Prove that $f \in L^p(\Omega, d\mu)$ for all $q \leq p < +\infty$ and

$$\lim_{p \rightarrow +\infty} \|f\|_p = \|f\|_\infty.$$

Solution.

Problem 4.8

Let (Ω, μ) and (Γ, ν) be two finite measure spaces and let $f \in L^1(\mu \times \nu)$. Show that if $f \geq 0$ and $1 \leq p < \infty$ then

$$\left(\int_\Omega \left(\int_\Gamma f(x, y) \nu(dy) \right)^p \mu(dx) \right)^{\frac{1}{p}} \leq \int_\Gamma \left(\int_\Omega f(x, y)^p \mu(dx) \right)^{\frac{1}{p}} \nu(dy).$$

Hint: Set $H(x) := \int_\Gamma f(x, y) \nu(dy)$ and check that

$$\|H\|_{L^p(\mu)}^p = \int_\Gamma \left(\int_\Omega f(x, y) H^{p-1}(x) \mu(dx) \right) \nu(dy).$$

Solution.

Problem 4.9

Assume that μ is the d -dimensional Lebesgue measure on $\mathbb{R}^d$, $1 \leq p < +\infty$ and let $f \in L^\infty(\mathbb{R}^d, d\mu)$ be such that $f \geq 0$ and f vanishes outside a ball B_R of radius $R > 0$. Let $j \in L^1(\mathbb{R}^d, d\mu)$ be such that $\int_{\mathbb{R}^d} j \, dx = 1$. Define,

$$j_\varepsilon(x) := \frac{1}{\varepsilon^d} j\left(\frac{x}{\varepsilon}\right) \quad \text{and} \quad f_\varepsilon = j_\varepsilon * f.$$

(a) Prove that if $p \geq 2$

$$|f_\varepsilon(x) - f(x)|^p \leq [\|f\|_\infty(1 + \|j\|_1)]^{p-2} |f_\varepsilon(x) - f(x)|^2.$$

(b) Use Hölder's inequality to prove that if $1 < p \leq 2$

$$\|f_\varepsilon - f\|_p \leq |B_R|^{\frac{1}{p}-\frac{1}{2}} \|f_\varepsilon - f\|_2.$$

(c) Deduce that if the approximation by C^∞ -functions theorem holds in $L^2(\mathbb{R}^d, d\mu)$ then it holds in $L^p(\mathbb{R}^d, d\mu)$.

Solution.